

An e-commerce platform inspired by Jarir Bookstore. Customers browse products across store branches, add them to a cart, and complete a purchase by placing an order and paying online.

USER

Personal and delivery info.
Owns one Cart, a history of Orders, and saved Cards.

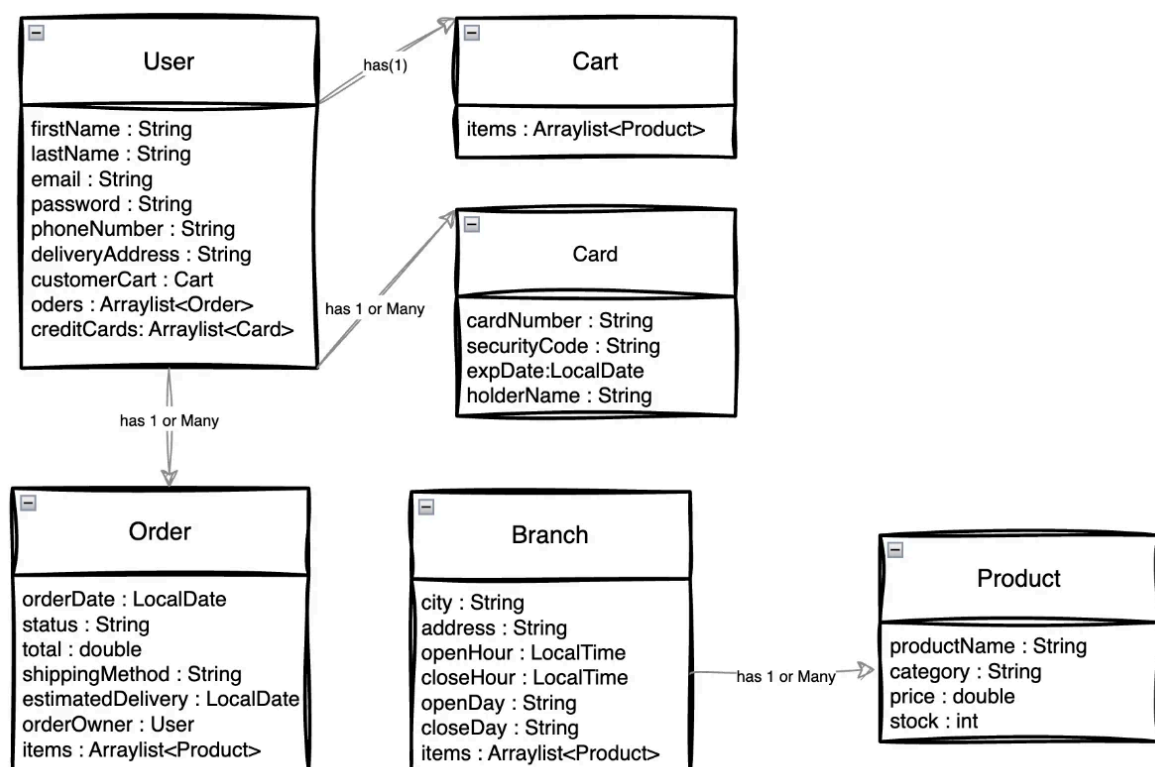
PRODUCT & BRANCH

Each item carries name, category, price and stock, and is linked to the branch that holds it.

ORDER

Created at checkout with the ordered products, total cost, shipping details and status.

MODELS



VALIDATIONS

Every constraint below is considered a model. Messages are returned if something went wrong.

User Model

User.java

```
1  @NotEmpty(message = "First Name Cannot be EMPTY !")
2  @Size(min=2,max=15, message = "First Name should be between 2 to 15")
3  @Pattern(regexp = "[a-zA-Z]+$", message = "Name must contain only letters (no
   numbers or Special Chars)")
4  private String firstName;
5
6  @NotEmpty(message = "Last Name Cannot be EMPTY !")
7  @Size(min=2,max=15, message = "Last Name should be between 2 to 15")
8  @Pattern(regexp = "[a-zA-Z]+$", message = "last Name must contain only letters
   (no numbers or Special Chars)")
9  private String lastName;
10
11 @NotEmpty(message = "Email Cannot be EMPTY !")
12 @Email(message = "Your email is invalid")
13 private String email;
14
15 @NotEmpty(message = "Phone number Cannot be EMPTY !")
16 @Pattern(regexp = "^\\d{10}$", message = "Phone number must be exactly 10 digits
   and contain only numbers")
17 private String phoneNumber;
18
19 @NotEmpty(message = "Password cannot be empty!")
20 @Size(min = 8, message = "Password must be at least 8 characters long")
21 @Pattern(regexp = "^(?=.*[a-z])(?=.*[A-Z])(?=.*[^a-zA-Z0-9]).+$", message =
   "Password must contain at least one uppercase letter, one lowercase letter, and
   one special character")
22 private String password;
23
24 @NotEmpty(message = "Address Cannot be EMPTY !")
25 private String deliveryAddress;
```

Order Model

Order.java

```
1  @NotNull(message = "orderDate Cannot be EMPTY !")
2  @FutureOrPresent(message = "The order date must be today or in the future.")
3  private LocalDate orderDate;
4
5  @NotEmpty(message = "status Cannot be EMPTY !")
6  @Pattern(regexp = "prepared|shipped|delivered", message = "Order status should be
   prepared OR shipped OR delivered")
7  private String status;
8
9  @NotNull(message = "total cannot be empty")
10 @Positive(message = "Total should be more than 0 ")
11 private Double total;
12
13 @NotEmpty(message = "shipping method Cannot be EMPTY !")
14 @Pattern(regexp = "pickUpFromBranch|delivery ", message = "choose
   pickUpFromBranch OR delivery ")
15 private String shippingMethod;
16
17 @NotNull(message = " estimated delivery Cannot be EMPTY !")
18 @FutureOrPresent(message = "estimated delivery date must be today or in the
   future.")
19 private LocalDate estimatedDelivery;
```

Branch Model

Branch.java

```
1  @NotEmpty(message = "city Cannot be EMPTY !")
2  @Size(min=2 ,message = "city more than 2 letters")
3  @Pattern(regexp = "[a-zA-Z]+$", message = "city must contain only letters (no
   numbers or Special Chars)")
4  private String city;
5
6  @NotEmpty(message = "Address Cannot be EMPTY !")
7  private String address;
8
9  @NotNull(message = "Open Hour Cannot be EMPTY !")
10 @JsonFormat(shape = JsonFormat.Shape.STRING, pattern = "HH:mm:ss")
11 private LocalTime openHour;
12
13 @NotNull(message = "Close Hour Cannot be EMPTY !")
14 @JsonFormat(shape = JsonFormat.Shape.STRING, pattern = "HH:mm:ss")
15 private LocalTime closeHour;
16
17 @NotEmpty(message = "Open Day Cannot be EMPTY !")
18 @Pattern(regexp = "Monday|Tuesday|Wednesday|Thursday|Friday|Saturday|Sunday",
   message = " Open Day should be from monday To sunday ")
19 private String openDay;
20
21 @NotEmpty(message = " Open Day Cannot be EMPTY !")
22 @Pattern(regexp = "Monday|Tuesday|Wednesday|Thursday|Friday|Saturday|Sunday",
   message = " Open Day should be from monday To sunday")
23 private String closeDay;
```

Product Model

Product.java

```
1  @NotEmpty(message = "Product Name Cannot be EMPTY !")
2  @Size(min=2, message = " Product Name more than 2 letters")
3  private String productName;
4
5  @NotEmpty(message = "you have to categorize the product")
6  @Pattern(regexp="Electronics|Computers|Books|VideoGames|OfficeSupplies|SchoolSupp
    lies|Toys", message = " choose from existing cetegories")
7  private String category;
8
9  @NotNull(message = "price cannot be empty")
10 @Positive(message = " price should be more than 0 ")
11 private Double price;
12
13 @NotNull(message = "stock cannot be empty")
14 @PositiveOrZero(message = " stock should be more than 0 ")
15 private Integer stock;
```

Card Model

Card.java

```
1  @NotEmpty(message = "card number Cannot be EMPTY !")
2  @Size(min = 16, max = 19, message = "The number must be between 16 and 19 digits
    long.")
3  @Pattern(regexp = "^[0-9]*$", message = "card number must contain digits only.")
4  private String cardNumber;
5
6  @NotEmpty(message = "CVV Cannot be EMPTY !")
7  @Size(min = 3, max = 3, message = "Security Code must be 3 digits only.")
8  @Pattern(regexp = "^\\d+$", message = "card CVV must contain digits only.")
9  private String securityCode;
10
11 @NotNull(message = "Expire Date Cannot be EMPTY !")
12 @FutureOrPresent(message = "Expire Date date must be today or in the future.")
13 private LocalDate expDate;
14
15 @NotEmpty(message = "Holder Name Cannot be EMPTY !")
16 @Size(min=2, message = "Holder Name should be more than 2 letters")
17 @Pattern(regexp = "^[a-zA-Z ]+$" ,message = "holder Name must contain only
    letters (no numbers or Special Chars)")
18 private String holderName;
```